INTERNET PROGRAMMING LAB MANUAL

Part-A

1. **Create an HTML5 page using tags to accomplish the following:**
   a. A paragraph containing text **"All that glitters is not gold".** Bold face and italicize this
   b. Create equation: **$x = 1/3(y^2 + z^2)$.**
   c. Put a background image to a page and demonstrate all attributes of background image
   d. Create unordered list of 5 fruits and ordered list of 3 flowers
2. **Create an HTML5 page that include three divisions and each division should implement background properties with below attributes:**
   a. **1st div** – Background Colour.
   b. **2nd div** – Background Gradient Colour.
   c. **3rd div** – Background image.
3. **Create following table using HTML tags.** Properly align cells, give suitable cell padding and spacing, and apply background colour, bold and emphasis necessary text.

Department	Sem1	SubjectA
		SubjectB
		SubjectC
	Sem2	SubjectE
		SubjectF
		SubjectG
	Sem3	SubjectH
		SubjectI
		SubjectJ

4. **Demonstrate the following HTML5 Semantic tags-** <article>, <aside>, <details>, <figcaption>, <figure>, <footer>, <header>, <main>, <mark>, <section> for a webpage that gives information about travel experience.
5. **Implement Below Tasks:**
   a. Create a class called **income**, and make it a background color of **#0ff**.
   b. Create a class called **expenses**, and make it a background color of **#f0f**.
   c. Create a class called **profit**, and make it a background color of **#f00**.
   Throughout the document, any text that mentions income, expenses, or profit, attach the appropriate class to that piece of text.
   Further create following line of text in the same document:
   The current price is 50₹ and new price is 40₹
6. **Insert 2 images using <img> tag and apply below styles:**
   a. Make a rounded dotted edge
   b. Make the Circle image with a double line edge
7. **Create a responsive design using grid layout by inserting the 3 divisions.**
   For Desktop
   Responsive design
   For Mobile
   Responsive design
8. **Divide a webpage into two section and insert iframe tags for each section and load the external sites as given below:**
   a. Iframe-1 - google.com
   b. Iframe-2 - any YouTube video.
9. **Create webpage which consists of three divisions each division should implement shorthand properties for border, padding, margin respectively.**
10. **Implement all three CSS types (Inline, Internal and External) and illustrate the properties with respect to types style sheets**

Part-B

1. **Create JavaScript program that illustrate the working of our statements (console.log(), alert(), document.write()).**
2. **Write JavaScript program that finds the greatest of three inputted numbers.**
3. **Write JavaScript program that finds the length of the given string.**
4. **Write a JavaScript program that reads (using prompt) a number and calculate its factorial.**
5. **Write a JavaScript program that reads numbers and stores in an array and search the given number in an stored array using linear search.**
6. **Write a JavaScript program that implements simple calculator Interface and perform arithmetic operations (+, -, /,).**
7. **Write a JavaScript program (using dom) that increase and decrease the font size of the text when an appropriate button get click on it.**
8. **Write a JavaScript program using DOM loads the image of a flower, fruit and an animal with respect to clicking on the specified button.**
9. **Write a JavaScript program that changes the entered text in a text box to the upper case using onchange event of the textbox.**
10. **Write a JavaScript program using JS function that validate the HTML form controls like textbox, dropdown, text area, radio button and checkbox.**

Examination:

- Two Questions has to be given from the above list A and B (one each).
- Student has to write and execute both of the programs

Marks Distribution:

Criteria	Marks
Practical Proper	
Writing & Execution of Program one from PART-A & one from PART-B	Questions from The List
(Each 30 marks)	
Total	**60**
Viva	**20**
Total	**80**

Part-A

1. **Create an HTML5 page using tags to accomplish the following:**
   a. A paragraph containing text **"All that glitters is not gold".** Bold face and italicize this
   b. Create equation: **$x = 1/3(y^2 + z^2)$.**
   c. Put a background image to a page and demonstrate all attributes of background image
   d. Create unordered list of 5 fruits and ordered list of 3 flowers

```
<!doctype html>
<html>
<head>
<title> HTML5 example</title>
<style>
body{
background-image:url("flower.jpg");
background-repeat: no-repeat;
background-size:contain;
background-position:left;
background-attachment: fixed;
background-color:light blue;
}
</style>
</head>
<body>
<p><b><i> all that glitters are not gold</i></b></p>
<p><b>x=1/3(Y<sub>1</sub><sup>2</sup>+Z<sub>1</sub><sup>2</sup>)</b></p>
<h2> 5 FRUITS </h2>
<ul>
<li> Apple </li>
<li> Banana </li>
<li> Charry </li>
<li> Date </li>
<li> Papaya </li>
</ul>
<h2> 3FLOWERS <h2>
<ol>
<li> Rose </li>
<li> Jasmin </li>
<li> Lillly </li>
</ol>
</body>
</html>
```

Output :



2. **Create an HTML5 page that include three divisions and each division should implement background properties with below attributes:**
   a. **1st div** – Background Colour.
   b. **2nd div** – Background Gradient Colour.
   c. **3rd div** – Background image.

```
<!doctype html>
<html>
<head>
<title> Background properties</title>
<style>
div{
width:100%;
height:100px;
margin:5px;
color:white;
text-align:center;
line-height:150px;
font-size:100px;
border:10px dotted blue;
}
.div1{
background-color:light blue;
color:green;
}
.div2{
background:linear-gradient(to left,red,yellow);
color:black;
}
.div3{
background-image:url("flower.jpg");
background-size:contain;
background-repeat:no-repeat;
background-position:center;
color:red;
```

```
}
</style>
</head>
<body>
<div class="div1"> winners</div>
<div class="div2"> winners</div>
<div class="div3"> winners</div>
</body>
</html>
```

Output



3.Create following table using HTML tags. Properly align cells, give suitable cell padding and spacing, and apply background colour, bold and emphasis necessary text.

Department	Sem1	SubjectA
		SubjectB
		SubjectC
	Sem2	SubjectE
		SubjectF
		SubjectG
	Sem3	SubjectH
		SubjectI
		SubjectJ

```
<!doctype html>
<html>
<head>
<title> department table</title>
</head>
<body>
<table border="2" cellpadding="10" cellspacing="1" width="700" align="center">
<tr>
<td rowspan="9" bgcolor="lightgray" align="center">
<b>Department</b></td>
<td rowspan="3" bgcolor="lightblue" align="center">
<b> Sem1</b></td>
<td bgcolor="lightyellow" align="center"><en>SubA</em></td>
</tr>
<tr>
<td bgcolor="lightyellow" align="center"><en>SubB</em></td>
</tr>
<tr>
<td bgcolor="lightyellow" align="center"><en>SubC</em></td>
```

```
</tr>
<tr>
<td rowspan="3" bgcolor="lightgreen" align="center">
<b> Sem2</b></td>
<td bgcolor="beige" align="center"><en>SubD</em></td>
</tr>
<tr>
<td bgcolor="beige" align="center"><en>SubE</em></td>
</tr>
<tr>
<td bgcolor="beige" align="center"><en>SubF</em></td>
</tr>
<tr>
<td rowspan="3" bgcolor="lightpink" align="center">
<b> Sem3</b></td>
<td bgcolor="pink" align="center"><en>SubG</em></td>
</tr>
<tr>
<td bgcolor="pink" align="center"><en>SubH</em></td>
</tr>
<tr>
<td bgcolor="pink" align="center"><en>SubI</em></td>
</tr>
</table>
</body>
</html>
```

Output:

Department	**Sem1**	SubA
		SubB
		SubC
	Sem2	SubD
		SubE
		SubF
	Sem3	SubG
		SubH
		SubI

4. **Demonstrate the following HTML5 Semantic tags-** <article>, <aside>, <details>, <figcaption>, <figure>, <footer>, <header>, <main>, <mark>, <section> for a webpage that gives information about ++travel experience.

```
<!doctype html>
<html>
<head>
<title> SEMANTIC TAGS</title>
<style>
body{
min-height:100vh;
```

```
display:flex;
flex-direction:column;
margin:0;
}
main{
flex:1;
padding:20px;
}
section{
background-color:lightblue;
padding:20pz;
}
footer{
background-color:lightgray;
text-align:center;
padding:15px;
}
</style>
</head>
<body>
<header><h1>MY TRAVEL WEBSITE</h1></header>
<main>
<section><h2>MY TRAVEL EXPERIANCE</h2>
<article><h3>TRIP TO OOTY</h3>
<p>OOTY IS<mark> BEAUTIFUL HILL STATION</mark></p>
<figure><img src="ooty.png" width="250">
<figcaption>BEAUTIFUL VIEW OF OOTY <figcaption></figure>
<details><summary> READ MORE </summary>
<p> OOTY IS FAMOUS FOR ITS MOUNTAINS,HILLS,GARDENS</P>
</details>
</article>
</section>
<aside>
<h3>TRAVEL TIPS</h3>
<p> CARRY WARM CLOTHES</p>
</aside>
</main>
<footer><p>  &copy;2026</p>
</body>
</html>
```

Output



5. **Implement Below Tasks:**
   a. Create a class called **income**, and make it a background color of **#0ff**.
   b. Create a class called **expenses**, and make it a background color of **#f0f**.
   c. Create a class called **profit**, and make it a background color of **#f00**.
   Throughout the document, any text that mentions income, expenses, or profit, attach the appropriate class to that piece of text.
   Further create following line of text in the same document:
   The current price is 50₹ and new price is 40₹

```
<!doctype html>
<html>
<head>
<style>
.income{background:#0ff;}
.expenses{background:#f0f;}
.profit{background:#f00;}
</style>
</head>
<body>
<p> The <span class="income">income</span> is more than <span class="expenses">expenses</span>, so there <span class="profit"> profit </span></p>
<p> The current price is <del> 50Rs </del> and new price is <ins> 40Rs </ins></p>
</body>
</html>
```

Output

The income is more than expenses, so there profit

The current price is ~~50Rs~~ and new price is 40Rs

6 Insert 2 images using <img> tag and apply below styles:

a. Make a rounded dotted edge

b. Make the Circle image with a double line edge

```
<!doctype html>
<html>
<head>
<style>
img{
width:180px;
height:180px;
margin:10px:
}
.round{
border:5px dotted red;
border-radius:10px;
}
.circle{
border:5px double blue;
border-radius:1000%;
}
</style>
</head>
<body>
<h5>  squre image <h5>
<img src="flower.jpg" class="round" alt="flower"><br>
<h5>  circle image <h5>
<img src="flower.jpg" class="circle" alt="flower">
</body>
</html>
```

Output

squre image



circle image



7 Create a responsive design using grid layout by inserting the 3 divisions.
For Desktop and **Responsive design**

```
<!doctype html>
<html>
<head>
  <title>Responsive Design</title>

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <style>
    body {
      font-family: Arial;
      text-align: center;
    }

    .container {
      display: flex;
      gap: 20px;
    }

    .box {
      background-color: lightblue;
      padding: 30px;
      flex: 1;
    }

    @media (max-width: 600px) {
      .container {
        flex-direction: column;
      }
    }
  </style>
</head>

<body>

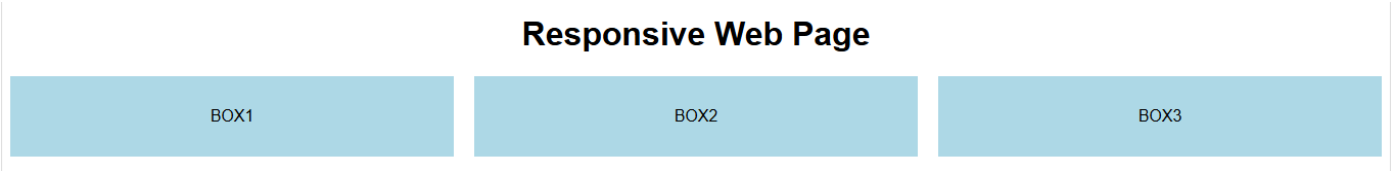
  <h1>Responsive Web Page</h1>

  <div class="container">
    <div class="box">BOX1</div>
    <div class="box">BOX2</div>
    <div class="box">BOX3</div>
  </div>

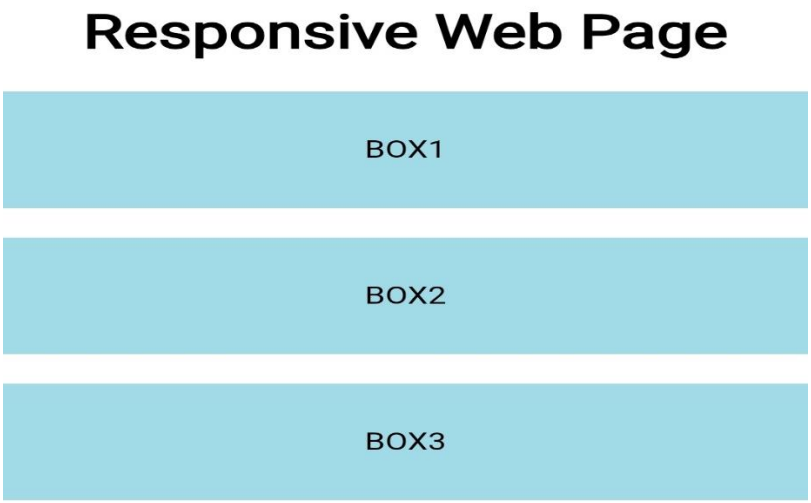
</body>
</html>
```

Output

For Desktop



For Mobile

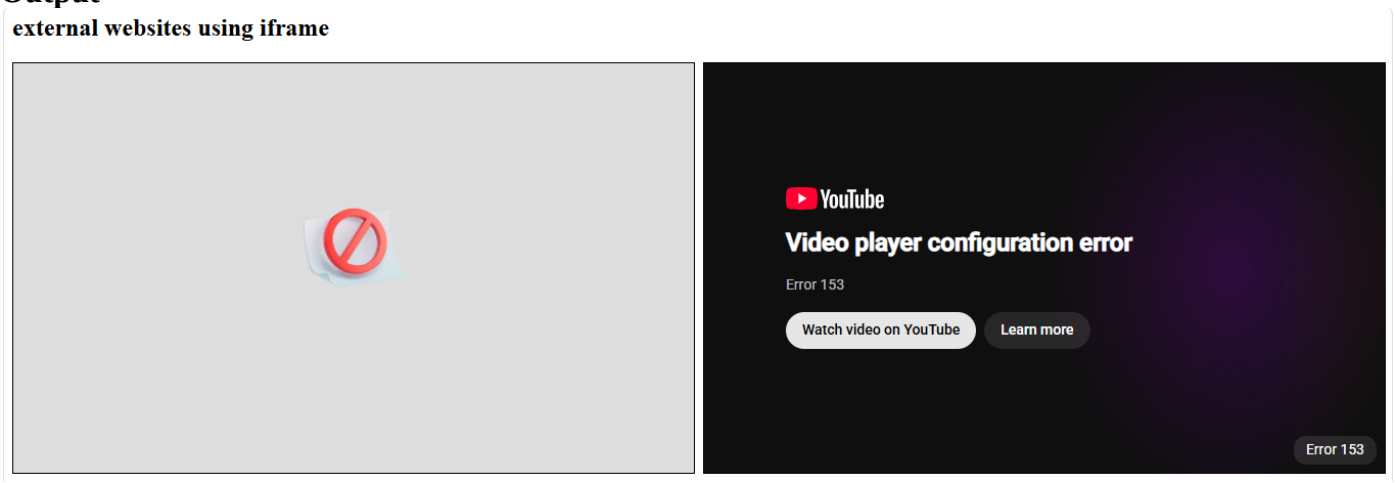


8. **Divide a webpage into two section and insert iframe tags for each section and load the external sites as given below:**
   a. Iframe-1 - google.com
   b. Iframe-2 - any YouTube video.

```
<doctype html>
<html>
<head>
<title>iframe exmple</title>
<style>
.container{
display:flex;
gap:10px;
}
.section{
width:50%;
height:400px;
}
iframe{
width:100%;
height:100%;
border:1px solid black;
}
</style>
</head>
<body>
<h2>external websites using iframe</h2>
<div class="container">
<div class="section">
<iframe src="https://www.google.com"></iframe>
</div>
<div class ="section">
```

```
<iframe src="https://www.youtube.com/embed/n6f1He8BtDQ" allowfullscreen>
</iframe>
</div>
</div>
</body>
</html>
```

Output

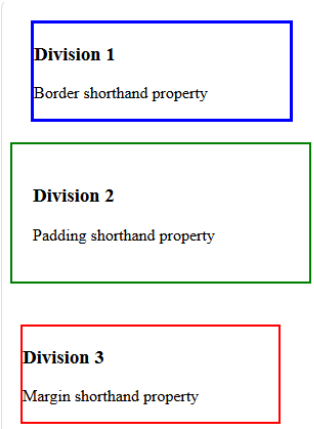


9. Create webpage which consists of three divisions each division should implement shorthand properties for border, padding, margin respectively.

```
<!doctype html>
<html>
<head>
<title>CSS shorthand properties</title>
<style>
.box1
{
   border: 3px solid blue;
   width: 250px;
   margin: 20px;
}
.box2
{
   border: 2px solid green;
   width: 250px;
   padding: 20px;
}
.box3
```

```
{
   border: 2px solid red;
   width: 250px;
   margin: 40px 30px 20px 10px;
}
</style>
</head>
<body>
<div class="box1">
   <h3>Division 1</h3>
   <p>Border shorthand property</p>
</div>
<div class="box2">
   <h3>Division 2</h3>
   <p>Padding shorthand property</p>
</div>
<div class="box3">
   <h3>Division 3</h3>
   <p>Margin shorthand property</p>
</div>
</body>
</html>
```

Output



10. Implement all three CSS types (Inline, Internal and External) and illustrate the properties with respect to types style sheets

```
<!doctype html>
```

```
<html>

<head>
   <title>Three Types of CSS</title>

   <!-- Internal CSS -->
   <style>
      h2 {
         color: purple;
         background-color: green;
         text-align: center;
         font-size: 28px;
         font-family: Arial;
         margin: 20px;
         padding: 10px;
         border: 3px solid black;
      }

      p {
         color: blue;
         font-size: 18px;
         font-style: italic;
         text-align: justify;
      }
   </style>

   <!-- External CSS -->
   <link rel="stylesheet" href="external.css">
</head>

<body>

   <!-- Inline CSS -->
   <h1 style="color: white;
            background-color: red;
            text-align: center;
            font-size: 20px;
            font-family: Verdana;
            margin: 20px;
            padding: 15px;
            border: 3px solid blue;">
      Inline CSS
   </h1>

   <!-- Internal CSS -->
   <h2>Internal CSS</h2>

   <p>
      This paragraph is styled by internal CSS
   </p>

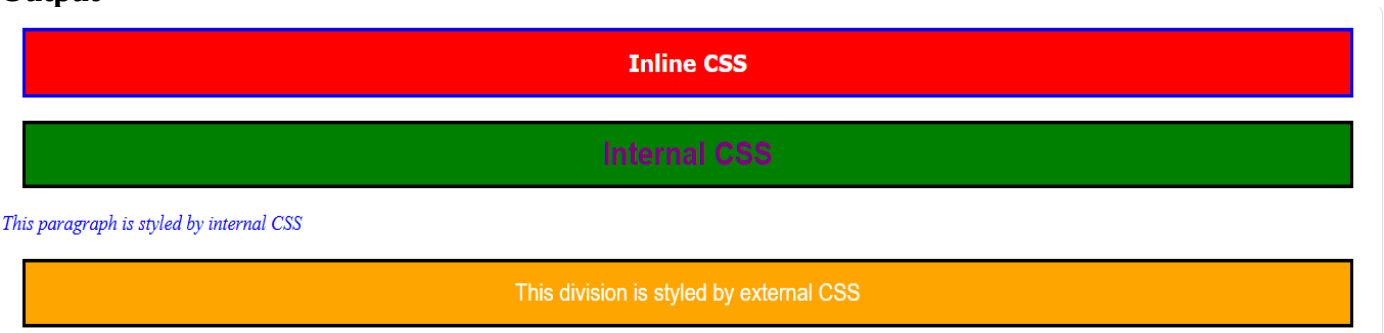
   <!-- External CSS -->
   <div>
      This division is styled by external CSS
   </div>
```

```
</body>
</html>
```

External.css

```
div {
   color: white;
   background-color: orange;
   text-align: center;
   font-size: 20px;
   font-family: Arial;
   margin: 20px;
   padding: 15px;
   border: 3px solid black;
}
```

Output

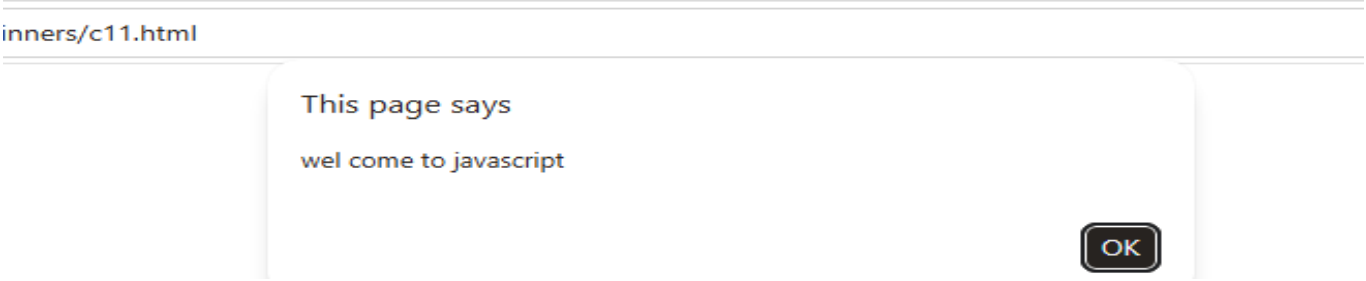


Part-B

1. **Create JavaScript program that illustrate the working of our statements (console.log(), alert(), document.write()).**

```
<!doctype html>
<html>
<head>
<title> Javascript output statements</title>
</head>
<body>
<script>
console.log("hello world");
alert("wel come to javascript");
document.write("This massage is displayed on webpage");
</script>
</body>
</html>
```

Output



This massage is displayed on webpage

2. **Write JavaScript program that finds the greatest of three inputted numbers.**

```
<!doctype html>
<html>
<head>
   <title>Greatest of Three Numbers</title>
</head>
<body>

<script>
let a = Number(prompt("Enter the first number"));
let b = Number(prompt("Enter the second number"));
let c = Number(prompt("Enter the third number"));

document.write("Entered numbers are: " + a + "  " + b + "  " + c);
document.write("<br>");

if (a >= b && a >= c)
{
   document.write(a + " is greatest");
}
else if (b >= a && b >= c)
{
   document.write(b + " is greatest");
}
else
{
   document.write(c + " is greatest");
}
</script>

</body>
</html>
```

Output

Entered numbers are: 7 9 10

10 is greatest

3. **Write JavaScript program that finds the length of the given string.**

```
<!doctype html>
<html>
<head>
   <title>Greatest of Three Numbers</title>
</head>
<body>

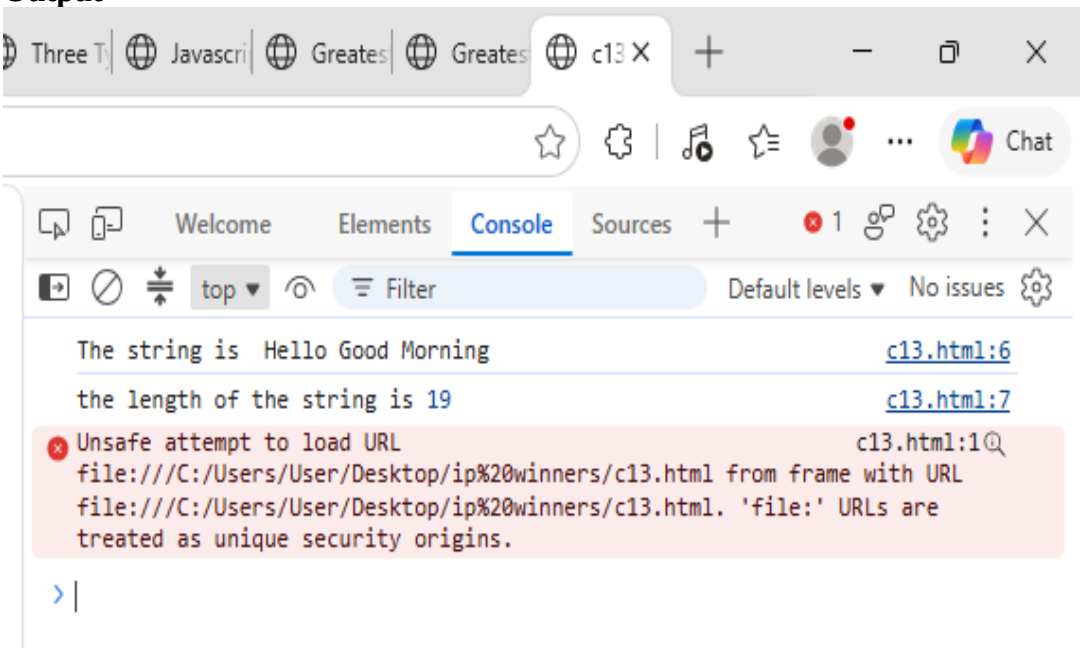
<script>
let a = Number(prompt("Enter the first number"));
let b = Number(prompt("Enter the second number"));
let c = Number(prompt("Enter the third number"));

document.write("Entered numbers are: " + a + "  " + b + "  " + c);
document.write("<br>");

if (a >= b && a >= c)
{
   document.write(a + " is greatest");
}
else if (b >= a && b >= c)
{
   document.write(b + " is greatest");
}
else
{
   document.write(c + " is greatest");
}
</script>

</body>
</html>
```

Output



4. **Write a JavaScript program that reads (using prompt) a number and calculate its factorial.**

```
<!doctype html>
<html>
<head>
<title> factorial </title>
</head>
<body>
<script>
let n=Number(prompt("enter the value"));
let fact=1;
for(i=1;i<=n;i++)
{
fact=fact*i;
}
document.write("factorial of ",n," is ",fact);
</script>
</body>
</html>
```

Output

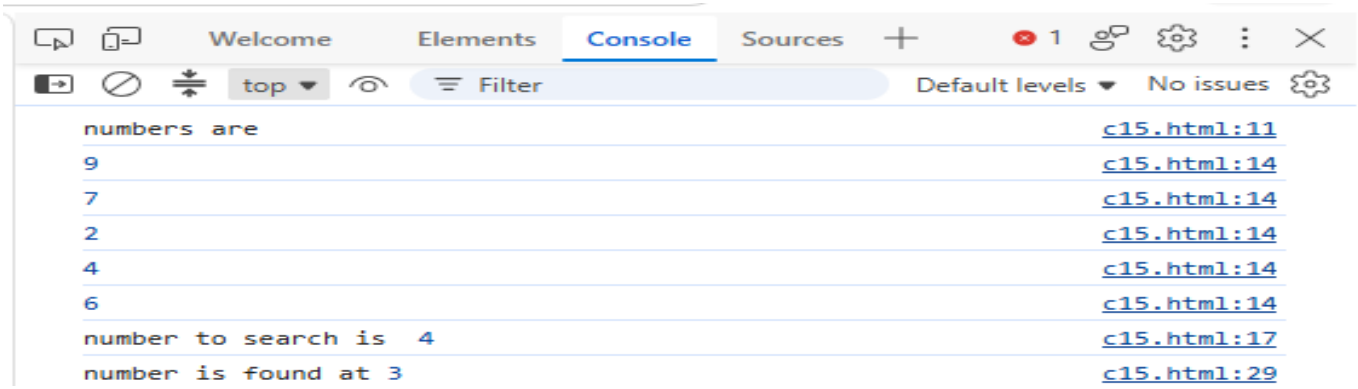
factorial of 5 is 120

5. **Write a JavaScript program that reads numbers and stores in an array and search the given number in an stored array using linear search.**

```
<!doctype html>
<html>
<body>
<script>
let a=[];
let n=Number(prompt("enter number of elements"));
for(let i=0;i<n;i++)
{
a[i]=Number(prompt("enter number"));
}
console.log("numbers are ");
for(let i=0;i<n;i++)
{
console.log(a[i]);
}
let key=Number(prompt("enter number to search"));
console.log("number to search is ",key);
let found=false;
let i;
for(i=0;i<n;i++)
{
if(a[i]==key)
{
found=true;
break;
}
}
```

```
if(found)
console.log("number is found at",i);
else
console.log("number is not found ");
</script>
</body>
</html>
```

Output



6. Write a JavaScript program that implements simple calculator Interface and perform arithmetic operations (+, -, /,).

```
<!doctype html>
<html>
<head>
<title>simple calculator</title>
</head>
</body>
<h2>simple calculator</h2>
<input type="number" id="num1"
placeholder="enter first number">
<br><br>

<input type="number" id="num2"
placeholder="enter second number">
<br><br>

<button onclick="calculate('+')">+</button>
<button onclick="calculate('-')">-</button>
<button onclick="calculate('*')">*</button>
<button onclick="calculate('/')">/</button>

<h3 id="result">result</h3>

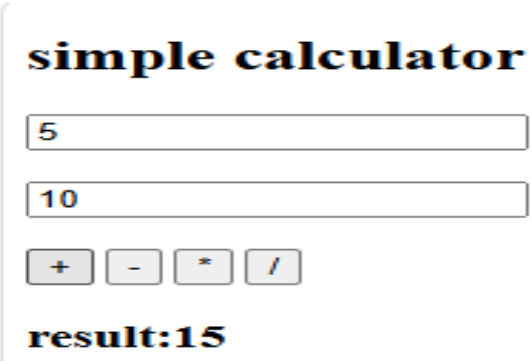
<script>
function calculate(operator)
{
let a=
Number(document.getElementById("num1").value);
let b=
Number(document.getElementById("num2").value);
let result;

if(operator == '+')
```

```
result=a+b;
 else if(operator == '-')
result=a-b;
else if(operator == '*')
result=a*b;
else if(operator == '/')
result=a/b;

document.getElementById("result").innerHTML="result:"+result;
}
</script>
</body>
</html>
```

OUTPUT



7. **Write a JavaScript program (using dom) that increase and decrease the font size of the text when an appropriate button get click on it.**

```
<!doctype html>
<html>
<head>
<title> font size change</title>
</head>
<body>
<p id="text">This is a sample text.</p>
<button onclick="increase()">increase</button>
<button onclick="decrease()">decrease</button>
<script>
let size=20;
function increase(){
size=size+2;
document.getElementById("text").style.fontSize=size +"px";
}
```

```
function decrease(){
size=size-2;
document.getElementById("text").style.fontSize=size +"px";
}
</script>
</body>
</html>
```

Output



8. **Write a JavaScript program using DOM loads the image of a flower, fruit and an animal with respect to clicking on the specified button.**

```
<!doctype html>
<html>
<head>
<title> image display using DOM</title>
</head>
<body>
<h2>Display image</h2>

<button onclick="showFlower()">Flower</button>
<button onclick="showFruit()">Fruit</button>
<button onclick="showAnimal()">Animal</button>
<br>
<br>
<img id="myImage" width="300" height="200">
<script>
function showFlower(){
document.getElementById("myImage").src="flowers.jpg";
}
function showFruit(){
```

```
document.getElementById("myImage").src="fruit.png";
}
function showAnimal(){
document.getElementById("myImage").src="animal.png";
}
</script>
</body>
</html>
```

Output



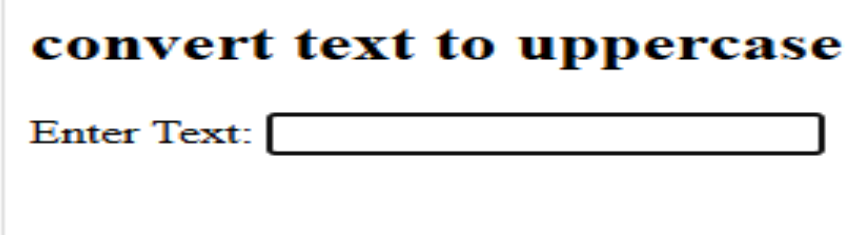
9. **Write a JavaScript program that changes the entered text in a text box to the upper case using onchange event of the textbox.**

```
<!doctype html>
<html>
<head>
<title>Uppercase conversion</title>
</head>
<body>
<h2>convert text to uppercase</h2>

Enter Text:
<input type="text" id="txt"
onchange="convertUppercase()">

<script>
function convertUppercase(){
var x=document.getElementById("txt");
x.value=x.value.toUpperCase();
}
</script>
</body>
</html>
```

Output



10. **Write a JavaScript program using JS function that validate the HTML form controls like textbox, dropdown, text area, radio button and checkbox.**

```
<!doctype html>
<html>
<head>
<title>Form Validation</title>

<script>
function validate()
{
   if(document.getElementById("name").value.trim() == "")
   {
      alert("Enter name");
      return false;
   }

   if(document.getElementById("city").value == "")
   {
      alert("Select city");
      return false;
   }

   if(document.getElementById("address").value.trim() == "")
   {
      alert("Enter address");
      return false;
   }
   if(!document.getElementById("male").checked &&
     !document.getElementById("female").checked)
   {
      alert("Select gender");
      return false;
   }

   if(!document.getElementById("terms").checked)
   {
      alert("Accept terms");
      return false;
   }
   alert("Form submitted successfully");
   return true;
}
</script>
</head>
<body>
<h2>Registration Form</h2>
```

```
<form onsubmit="return validate( )">
   Name:
   <input type="text" id="name">
   <br><br>
   City:
   <select id="city">
      <option value="">Select City</option>
      <option value="Davangere">Davangere</option>
      <option value="Bangalore">Bangalore</option>
      <option value="Mysore">Mysore</option>
   </select>
   <br><br>
   Address:
   <textarea id="address"></textarea>
   <br><br>
   Gender:
   <input type="radio" id="male" name="gender"> Male
   <input type="radio" id="female" name="gender"> Female
   <br><br>
   <input type="checkbox" id="terms">
   I accept the terms and conditions
   <br><br>
   <input type="submit" value="Submit">
</form>
</body>
</html>
```

Output

